

AE-499: Stocksolution of $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNP}\{\text{tBu}\})]$ in water

Date: 2025-05-21

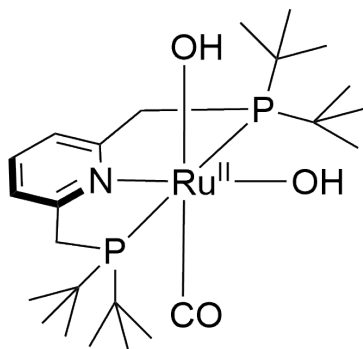
Tags: PNP Degassed Stocksolution
 $\text{P}(\text{tBu})\text{N}(\text{py})\text{P}(\text{tBu})$ $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNP})]$
AE

Category: Two-photon

Status: Done

Created by: Alexander Eith

Reaction scheme/sample structure



Chemical Formula: $\text{C}_{24}\text{H}_{45}\text{NO}_3\text{P}_2\text{Ru}$

Molecular Weight: 558,64552

ChemDraw file: [AE-493.cdxml](#)

Literature/reference experiments

Literature	/
Reproduction	/
Related experiment	Two-photon - AE-486: Stock solutions of $[\text{Ru}(\text{CO})(\text{PNP})(\text{OH})_2]$ and KC-098

Reagents

Name	CAS/Experiment number	Amount $[\mu\text{mol}]$	Equivalents	Mass _{theo} [mg]	Mass _{exp} [mg]	Molar mass [g/mol]	Volume [ml]	Density [g/mL]
$[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNP}\{\text{tBu}\})]$	Two-photon - AE-456: Synthesis of $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNP}\{\text{tBu}\})]$ using Ag_2O, 5 h	20	/	11.17	11.34	558.65	/	/
Milli-Q water	Prep work - AE-442: Degassing of Milli-Q water	/	/	/	/	/	10	0.998

Procedure/observations

All steps were done according to [\[Protocol\] Schlenk Technique](#).

All steps were carried out in the dark.

Date	Time	Step	Observations	Pictures
22.05	13:55	A 20 mL Schlenk flask was set under argon.	/	/
	14:15	[Ru] was added in air	/	/
		The flask was set under argon	/	/
	14:35	To the flask water (10 mL) was added	yellow-orange solution	start of stirring.jpg
	14:40	The mixture was stirred at 500 rpm	no change	end of stirring.jpg
		Another 20 mL schlenk flask was prepared	/	/
	15:45	The mixture was transferred into the new schlenk flask and thereby filtered thorough a PA syringe filter (pore diameter = 0.2 µm)	significantly lighter in tone, hue did not change	after filt.jpg
	15:55	The flask was stored at -20 °C in the dark under argon in Equipment - Freezer Lab E004 CEEC II	AE-499-1	/

Product characterization and results

Sample	Volume [mL]	Purity	Concentration [mM]	Description	Storage location
AE-499-1	10	>98 %	concentrated solution (approx. 1.44)	light yellow solution	Equipment - Freezer Lab E004 CEEC II

Linked experiments

- [AE-243: 1 M Stock solution of KOH in water](#)

Two-photon - [AE-486: Stock solutions of \[Ru\(CO\)\(PNP\)\(OH\)₂\] and KC-098](#)

Linked resources

Equipment - [Freezer Lab E004 CEEC II](#)

Protocol - [Schlenk Technique](#)

Protocol - [Degassing of fluids in normal Schlenk flask](#)

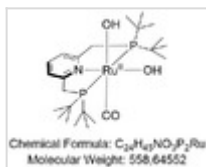
Attached files

AE-493.cdxml

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AE-493.png

sha256: 7afc615e0924e2e4799e3f4445c5c7c48c993a83c9aacd201cc7bc395f6a1351



end-of-stirring.jpg

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start-of-stirring.jpg

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after-filt.jpg

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Comment

On 2025-06-03 11:19:33 Jacob Schneidewind wrote:

Notes:

* Adding product characterization table, specifying flask, location etc. there



Unique eLabID: 20250521-e881049bea9f430d720bc3e5e1ec7e4bc19a6261

Link: <https://elab.water-splitting.org/experiments.php?mode=view&id=2191>